

★ WEEK 14 · PART 1 of 3

Markov Chain Monte Carlo: Sample, Don't Integrate

The Engine That Made Bayes Practical

Monte Carlo · the histogram IS the posterior · why a Markov chain

Last week's catch was one intractable integral. MCMC sidesteps it entirely: instead of computing the posterior, we draw samples from it — and read every answer off the draws.

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PART 1 · SAMPLE, DON'T INTEGRATE

Turning Integrals Into Averages

The integral that blocked us, Monte Carlo turning it into an average of draws, the histogram as the posterior, and why a Markov chain is the way to generate those draws.

Born at Los Alamos — Part 1

An algorithm to dodge an impossible integral

In the late 1940s, physicists at Los Alamos faced integrals no one could solve. Stanislaw Ulam and John von Neumann answered with Monte Carlo — estimate the answer by random sampling — a method named for the famous casino.

The problem they faced

- Physics calculations needed impossible integrals.
- No formula, and grids were hopeless in high dimensions.
- They needed a completely different approach.

The Monte Carlo answer

- Estimate the integral by random sampling.
- Averaging draws approximates the quantity.
- The 1953 Metropolis paper made it a chain.

Born at Los Alamos — Part 2

From physics to the Bayesian revolution

The Metropolis method sat mostly in physics for decades. Hastings generalized it in 1970, and in the 1980s and 90s statisticians realized it was the key to computing posteriors — sparking the modern Bayesian revolution.

The long incubation

- For years it was a physicist's tool.
- Hastings generalized the algorithm in 1970.
- Its statistical power went unnoticed for a while.

The revolution

- It was exactly what Bayesian inference needed.
- Suddenly, real posteriors became computable.
- It is why Bayes is everywhere today.

The Integral That Blocked Us

Where Week 13 left off

1

Bayes needs a normalizer

The posterior requires dividing by the evidence — an integral of the likelihood times the prior over all parameters.

2

That integral is usually intractable

For any realistic model it has no closed form, and in many dimensions it can't be done numerically either.

3

MCMC sidesteps it entirely

Instead of computing the posterior, we draw samples from it — and every summary we want is just a summary of those draws.

Part 1 in one line: don't compute the posterior — sample it, and read the answers off the draws.

The Problem We Are Solving

One integral stood in the way

Everything hinges on the normalizing integral. Where conjugacy gave it for free, real models don't. Without it, the posterior is known only up to a constant — enough to evaluate, but not, it seems, to use.

What we can do

- Evaluate the likelihood times the prior at any point.
- So we know the posterior's SHAPE.
- We just don't know its scale (the normalizer).

What we can't do

- Compute that normalizing integral directly.
- So we can't turn the shape into a proper distribution.
- And we can't sample from it the easy way.

Monte Carlo: Integrals Become Averages

The move that changes everything

$$\mathbb{E}_p[f(\theta)] = \int f(\theta) p(\theta) d\theta \approx \frac{1}{N} \sum_{i=1}^N f(\theta_i), \quad \theta_i \sim p$$

The identity

- An expectation is an integral against a distribution.
- It also equals the average of samples from that distribution.
- So sampling replaces integrating.

Why it helps

- Averages are easy; high-dimensional integrals are not.
- The error shrinks predictably with more draws.
- It scales to dimensions where grids fail completely.

From Integrals to Averages

A hard analytic problem becomes an easy numerical one

The old way

- Set up and solve an integral analytically.
- Or grid the space and sum — hopeless in high dimensions.
- Both fail for realistic posteriors.

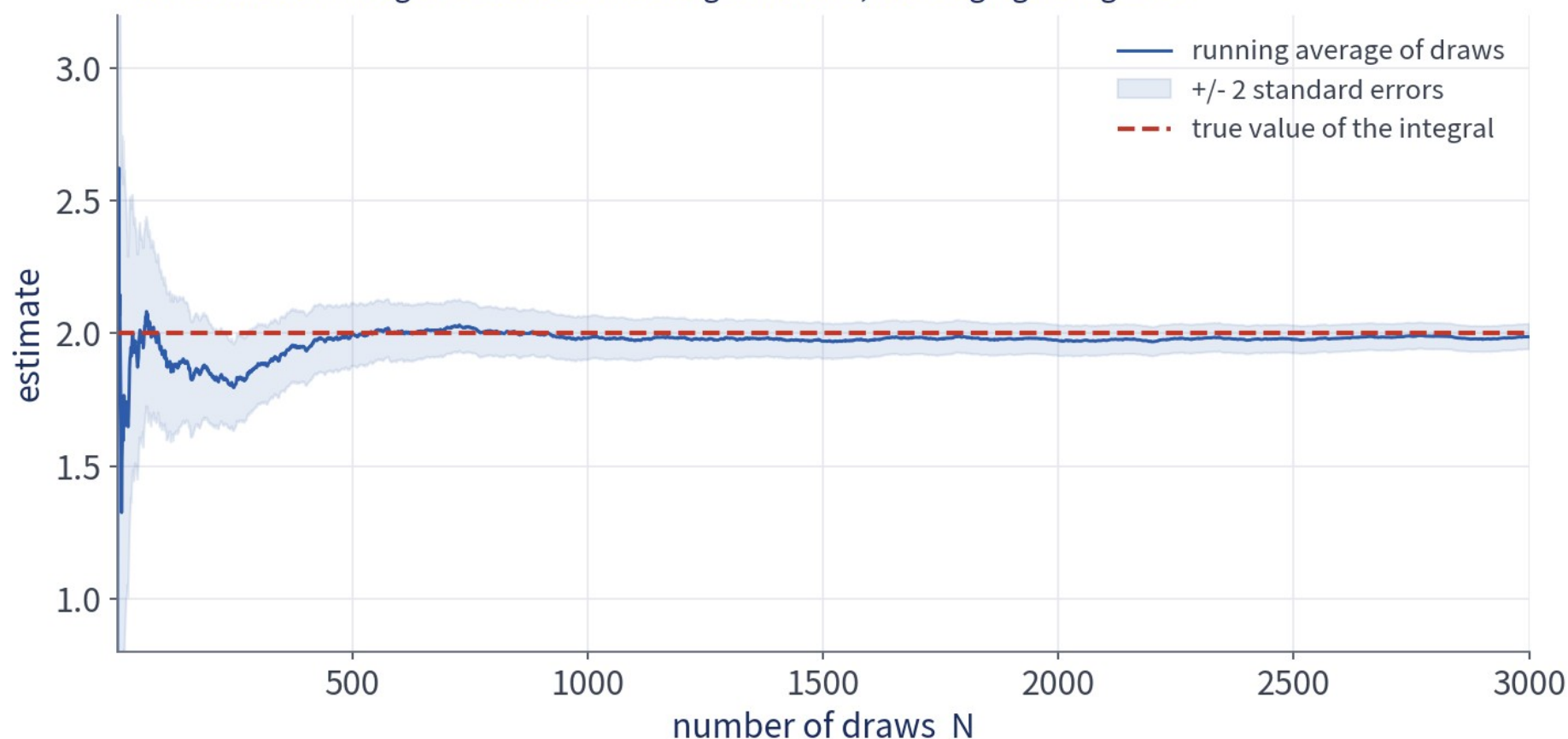
The Monte Carlo way

- Draw samples from the distribution.
- Average the quantity of interest over them.
- Accuracy improves with the square root of the draws.

An Integral, Estimated by Averaging

The running average of draws converges to the true value of the integral, and its error band narrows as the number of draws grows. Integration has become simple averaging.

Monte Carlo: an integral becomes an average of draws, converging as N grows



Sample, Then Summarize

The whole plan in four words

The plan is simple: get a big pile of draws from the posterior, then compute whatever you want as a summary of those draws. Every Bayesian quantity — means, intervals, forecasts — is just a statistic of the samples.

Step one: sample

- Generate many draws from the posterior.
- Each draw is a plausible parameter value.
- Together they represent the whole distribution.

Step two: summarize

- The mean of the draws is the posterior mean.
- Their quantiles give a credible interval.
- Their spread is the uncertainty.

From Samples to Answers

Everything you want is a summary of draws

Point summaries

- Posterior mean: average the draws.
- Posterior median or mode: read off the draws.
- Any function of the parameter: transform each draw.

Uncertainty and prediction

- Credible interval: take sample quantiles.
- Probability of a claim: fraction of draws satisfying it.
- Predictive: simulate new data from each draw.

The Histogram IS the Posterior

Collect draws; read every answer off them

There is no separate 'posterior' to compute — the histogram of your draws IS the posterior, as far as any practical question is concerned. With enough samples, the histogram matches the true density to any precision you like.

What this means

- The draws stand in for the distribution itself.
- Any summary of the density is a summary of the draws.
- You never need the closed-form density.

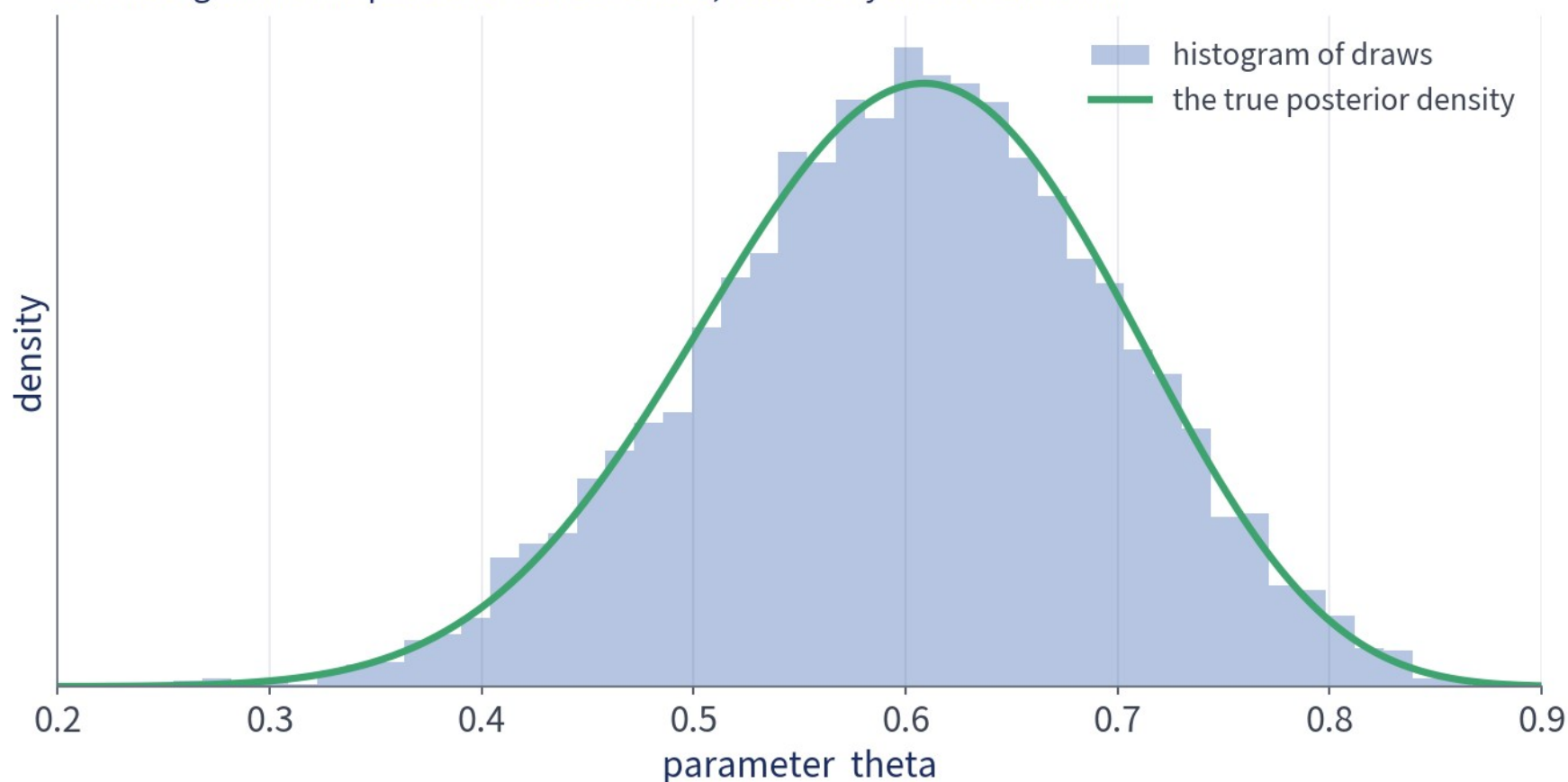
The reassurance

- More draws, a closer match to the truth.
- The approximation error is controllable.
- It is exact in the limit of infinite draws.

The Histogram IS the Posterior

The histogram of a few thousand draws traces the true posterior density almost exactly. Once you have the draws, the density itself is redundant — the samples answer every question.

The histogram IS the posterior: collect draws, read every answer off them



But We Can't Sample It Directly

The posterior isn't a distribution we can just draw from

Why not

- We only know the posterior up to the normalizer.
- It is not a standard distribution with a built-in sampler.
- It lives in many dimensions with a complex shape.

The bind

- Monte Carlo needs draws from the posterior.
- But we can't draw from it independently.
- We need a cleverer way to generate samples.

We CAN Evaluate the Target

Even without the normalizer

$$\text{we CAN evaluate } p(\theta \mid \text{data}) \propto \underbrace{P(\text{data} \mid \theta) P(\theta)}_{\text{no normalizer needed}}$$

What we can compute

- At any parameter value, the likelihood times the prior.
- That is the posterior's shape, up to a constant.
- No integral is needed to evaluate it at a point.

Why this is enough

- The chain only ever compares two points.
- In a ratio, the unknown constant cancels.
- Pointwise evaluation is all MCMC requires.

Why a Markov Chain?

Build a walk that settles into the posterior

Since we can't draw from the posterior directly, we build a random walk that visits parameter values in proportion to the posterior. Let it run, and the values it visits become our samples.

The idea

- Construct a walk over the parameter space.
- Design it to linger where the posterior is high.
- Its long-run visits mirror the posterior.

Why a chain works

- Each step only needs the target at two points.
- So the normalizer never appears.
- The walk explores the whole distribution over time.

What a Markov Chain Is

We have met these before

The definition

- A sequence where the next step depends only on the current one.
- It is memoryless beyond the present (recall Week 10).
- Here the 'state' is a parameter value.

The new twist

- In Week 10 the chain was the model.
- Here the chain is a COMPUTATIONAL device.
- We design it so its long-run behavior is the posterior.

The Stationary Distribution

Make the walk's long-run law the posterior

design a chain whose stationary distribution is $\pi(\theta) = p(\theta \mid \text{data})$

The target

- Every chain has a long-run stationary distribution.
- We engineer the walk so that law IS the posterior.
- Then its visited values are posterior draws.

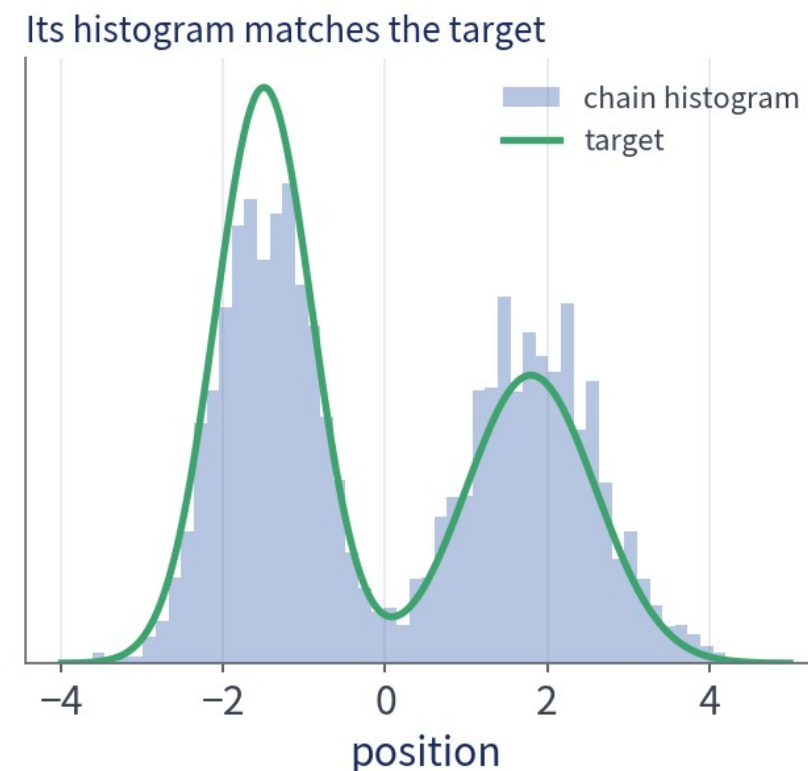
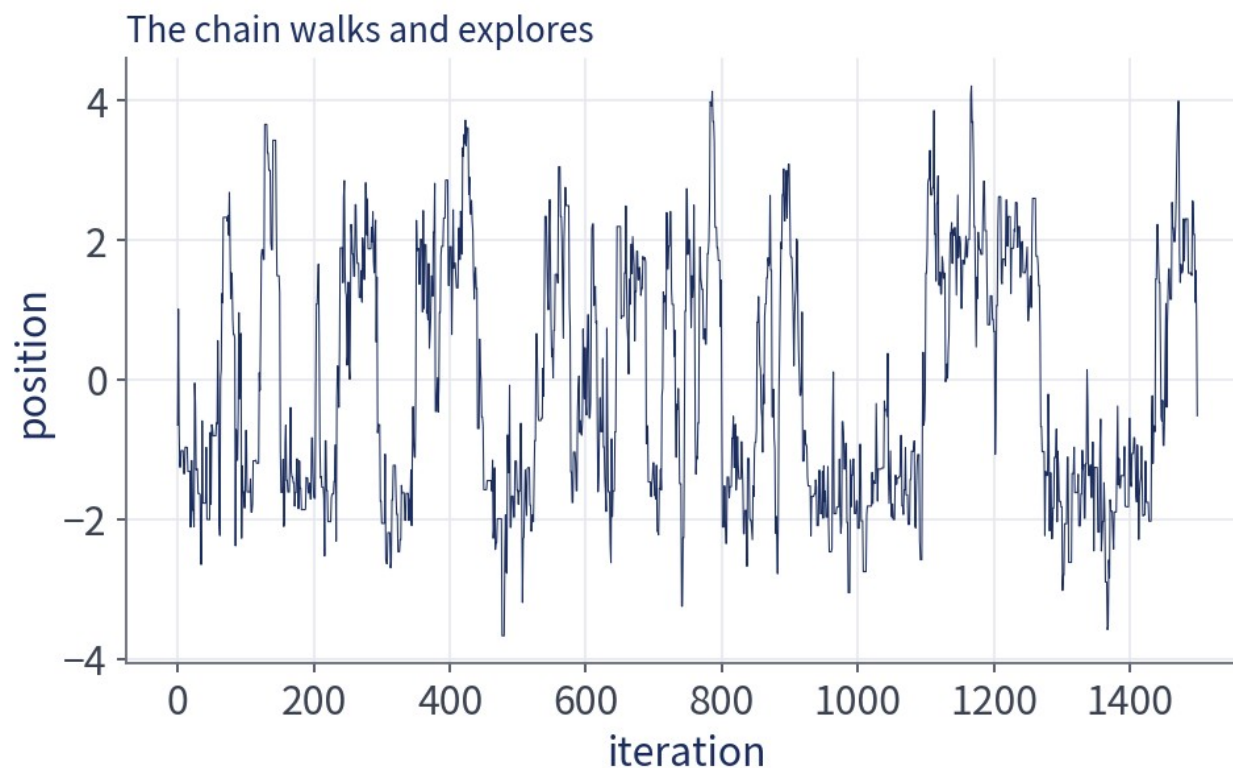
Why it works

- After it settles, positions are distributed as the posterior.
- Collect them, and you have your samples.
- The algorithms of Part 2 build exactly such walks.

A Chain Settling Into the Target

A well-designed walk explores the parameter space, and the histogram of where it goes matches the target distribution — even a two-humped one. The chain's visits become posterior samples.

A Markov chain, designed well, settles into the target distribution



Correlated but Valid Samples

MCMC draws are dependent — and that's okay

The dependence

- Each draw is a small step from the last.
- So consecutive samples are correlated, not independent.
- The chain wanders rather than jumping freely.

Why it still works

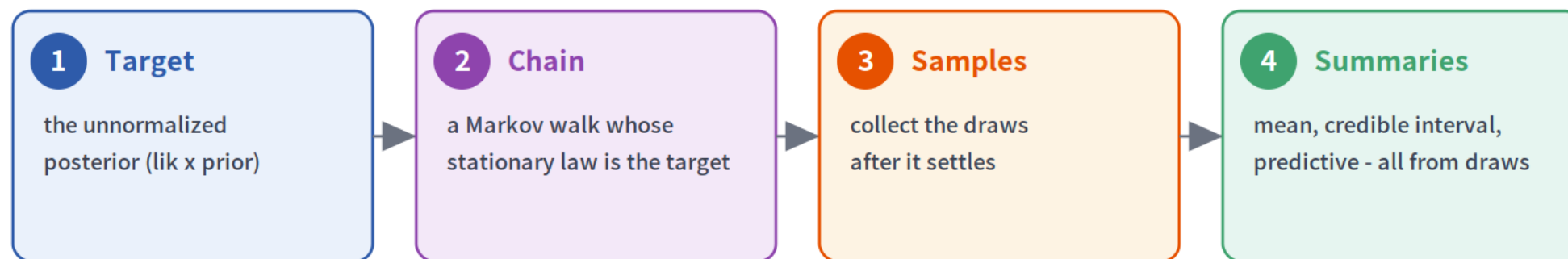
- Averages over the draws are still valid.
- You just need more draws for the same precision.
- Diagnostics (Part 3) tell you how many are enough.

The MCMC Recipe at a Glance

Take the unnormalized target, build a Markov chain whose stationary law is that target, collect the draws once it settles, and read all your summaries off them. Evaluate the target; never integrate it.

The MCMC Recipe at a Glance

Never compute the integral - build a walk that visits values in proportion to the posterior.



The whole trick in one line

You only ever need to EVALUATE the target (likelihood times prior) at points - never to integrate it.
The chain turns those pointwise evaluations into a stream of draws whose histogram is the posterior.

Sample, don't integrate - then read every answer off the draws.

What Parts 2 and 3 Add

From idea to working sampler

Part 2 — the algorithms

- Metropolis-Hastings: propose a move, accept or reject.
- Why the normalizer cancels in the acceptance rule.
- Gibbs sampling: update one parameter at a time.

Part 3 — the craft

- Trace plots and convergence diagnostics.
- Burn-in, mixing, and how many draws you need.
- Modern tools like PyMC, and the models we can now fit.

Monte Carlo Integration

Direct Claude Code to turn an integral into an average

GOAL See Monte Carlo in action: estimate a quantity by sampling, and watch the estimate converge.

[INPUT] Prompt · Claude Code

Write and run monte_carlo.py that:

1. picks a quantity with a known answer - e.g. estimate pi by sampling points in a square, or an expectation under a known distribution;
 2. estimates it as the average over N random draws;
 3. plots the running estimate vs N with an error band, and the true value as a line;
 4. shows the error shrinking like one over root N.
- Confirm the estimate converges to the known answer.

How to work with it

- An integral becomes an average of draws.
- The estimate converges as N grows.
- Error falls like one over root N.
- This is the engine under all of MCMC.

Deliverable: monte_carlo.py + a running-estimate plot converging to the known value.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

- | | | |
|-----------|---|-------------------------------------|
| Q1 | Monte Carlo replaces an integral with what? | An average of draws |
| Q2 | For MCMC, what do we need to compute at a point? | The target, up to a constant |
| Q3 | We design the Markov chain so its stationary law is what? | The posterior |

All three come from Monte Carlo, the unnormalized target, and the stationary-distribution design.

Three Misconceptions to Avoid

Where MCMC intuition slips

Common myths

- "MCMC computes the normalizing integral."
- "You need independent draws for valid answers."
- "MCMC finds the single best parameter."

The reality

- It sidesteps the integral — only ratios are needed.
- Correlated draws are fine; you just need more of them.
- It maps the WHOLE posterior, not one point.

Three Things to Carry Forward

Part 1 in one screen

1

Sample, don't integrate

Monte Carlo turns an intractable integral into an easy average of draws. Every posterior summary — mean, interval, forecast — becomes a summary of samples.

2

The histogram is the posterior

With enough draws, their histogram matches the posterior to any precision. We only ever need to EVALUATE the target (likelihood times prior), never to normalize it.

3

A Markov chain generates the draws

Since we can't sample the posterior directly, we build a walk whose stationary distribution IS the posterior. Its correlated-but-valid visits are our samples.

Next — Part 2: Metropolis-Hastings and Gibbs, the algorithms that build the chain.